WORKED SOLUTION

STEP III 2000, Question 12

Probability and Statistics • Poisson Approximation

STEP III 2000, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q12

The key idea

Once you know your own ticket wins, the only randomness left is in the other players' tickets. Each of them independently matches the winning number with probability $\frac{1}{N}$, so the number of *other* winners is binomial. With many players and a small success probability, that binomial is close to a Poisson distribution, and the given limit $\left(1-\frac{a}{n}\right)^n \approx \mathrm{e}^{-a}$ is exactly what makes this work.

Solution

The exact probability

Suppose you hold a winning ticket. Let X be the number of the other $2N-1$ players who also hold the winning number, so the total number of winners is $W = 1 + X$.

The machine chooses each player's number uniformly and independently of everyone else's, and independently of the draw. So, given the winning number (whatever it is) and given your ticket, each of the other $2N-1$ players matches it with probability $\frac{1}{N}$, independently of the others. Hence

$$X \sim \mathrm{B}\left(2N-1,\ \tfrac{1}{N}\right).$$

Three or fewer winners means $X \leq 2$, so the exact probability is

$$\mathrm{P}(W \leq 3) = \sum_{k=0}^{2} \binom{2N-1}{k} \left(\frac{1}{N}\right)^k \left(1-\frac{1}{N}\right)^{2N-1-k}.$$

Written out in full, this is

$$\left(1-\frac{1}{N}\right)^{2N-1} + \frac{2N-1}{N}\left(1-\frac{1}{N}\right)^{2N-2} + \frac{(2N-1)(2N-2)}{2N^2}\left(1-\frac{1}{N}\right)^{2N-3}.$$

The approximation

For each fixed $k \in \{0, 1, 2\}$, split the power:

$$\left(1 - \frac{1}{N}\right)^{2N-1-k} = \left[\left(1 - \frac{1}{N}\right)^{N}\right]^{2} \left(1 - \frac{1}{N}\right)^{-(1+k)}.$$

Using the given fact with $a = 1$ and $n = N$, the bracket is about e^{-1}. The last factor is a fixed small power of a number very close to 1, so it is about 1. Hence each power is about e^{-2}.

For the binomial coefficients, k is at most 2 and N is large, so

$$\binom{2N-1}{k} \frac{1}{N^k} = \frac{(2N-1)(2N-2)\cdots(2N-k)}{k!\, N^k} \approx \frac{(2N)^k}{k!\, N^k} = \frac{2^k}{k!}.$$

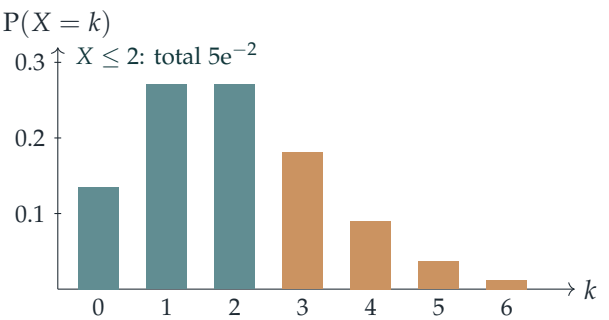
Putting these together, the terms for $k = 0, 1, 2$ are about e^{-2}, $2\mathrm{e}^{-2}$ and $2\mathrm{e}^{-2}$, so

$$\mathrm{P}(W \leq 3) \approx \mathrm{e}^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) = 5\mathrm{e}^{-2} \approx 0.677 \approx \frac{2}{3},$$

using $\mathrm{e}^2 \approx 7.39$. This is the required result.

Ask yourself Why does X behave like a Poisson variable with mean 2?

The calculation above is the standard Poisson limit of a binomial. The number of trials, $2N - 1$, is large, and the success probability $\frac{1}{N}$ is small, while their product stays close to 2. In that regime $\binom{n}{k} p^k (1-p)^{n-k} \to \mathrm{e}^{-\lambda}\lambda^k/k!$ with $\lambda = np$, and here $\lambda \to 2$.



The limiting distribution of X is Poisson with mean 2. The teal bars ($X \leq 2$, so at most three winners in all) make up about two-thirds of the total.

Players choosing their own numbers

If players choose, the numbers are no longer uniform: people favour birthdays (numbers up to 31), lucky numbers and patterns. Suppose a player picks number i with probability p_i, where $\sum p_i = 1$, and the winning number is still drawn uniformly. Then, given

that you win, your number is i with probability proportional to $p_i \cdot \frac{1}{N}$, that is, with probability p_i. So popular numbers are over-represented among winning tickets. If your number is i, the number of other winners is $\mathrm{B}(2N-1, p_i)$, with mean $(2N-1)p_i$. Averaging over i, the expected number of other winners is

$$(2N-1)\sum_i p_i^2 \;\geq\; (2N-1)\cdot\frac{1}{N},$$

by the Cauchy–Schwarz inequality $\left(\sum p_i\right)^2 \leq N\sum p_i^2$, with equality only when every $p_i = \frac{1}{N}$. So non-uniform choices mean more people sharing your win, on average, and the probability of three or fewer winners would typically *decrease*. (A single player who deliberately picks an unpopular number would do better than average, but players as a whole do worse.)

The expected number of winners

Since $X \sim \mathrm{B}\left(2N-1, \frac{1}{N}\right)$, its mean is $np = (2N-1)\cdot\frac{1}{N}$. (Equivalently, write X as a sum of $2N-1$ indicator variables, one for each other player, each with mean $\frac{1}{N}$.) Therefore

$$\mathrm{E}(W) = 1 + \mathrm{E}(X) = 1 + \frac{2N-1}{N} = 3 - \frac{1}{N} = 3 - N^{-1},$$

as required. Note this is exact, not an approximation.

Checks

- For $N = 2$ and $N = 3$, enumerating every assignment of numbers to the $2N$ players and every winning number gives exactly the binomial expression and exactly $3 - \frac{1}{N}$.
- The exact probability is 0.7054 for $N = 10$, 0.6794 for $N = 100$ and 0.6767 for $N = 10\,000$, approaching $5\mathrm{e}^{-2} = 0.67668$.
- A simulation with $N = 50$ (10^5 weeks) gave 0.683 against the exact 0.682, and a mean of 2.981 winners against $3 - \frac{1}{50} = 2.98$.
- With $N = 10$ and half the numbers twice as popular as the others, the probability falls from 0.705 to 0.641, and the expected number of winners rises from 2.9 to 3.11.

Common slips

- Using $2N$ trials instead of $2N-1$: your own ticket is already known to win, so only the other players are random. With $2N$ trials the expectation would come out as $1 + 2 = 3$, not $3 - \frac{1}{N}$.
- Using the unconditional mean: across all weeks the expected number of winners is $2N \cdot \frac{1}{N} = 2$, but knowing that you win adds one certain winner and leaves $2N-1$ others.
- Forgetting the $k = 2$ term, or writing it as $2^2\mathrm{e}^{-2}$ without the $\frac{1}{2!}$.

Beyond STEP

The approximation here is the law of rare events: a binomial with many trials and a small success probability converges to a Poisson distribution. The final part touches on the inspection paradox, or size-biased sampling. Conditioning on *your* ticket winning makes crowded numbers more likely, which is why the conditional mean is 3 rather than 2. Both ideas are treated well in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

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